Real Estate Investment Advisor

Project Documentation

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This document describes a machine learning project that (1) classifies whether a residential property is a Good Investment and (2) estimates its price after five years. Results are based on the india_housing_prices dataset (250,000 properties). Several results depend on rule-based targets and assumed growth rates, which are explained in Sections 1 and 4.

1. Methodology

This section describes how the data was cleaned, how features and targets were created, how the models were trained and evaluated, and how experiments were tracked.

1.1 Data source

- **Dataset:** `india_housing_prices.csv`
- **Size:** 250,000 rows and 23 columns in the original file
- **Columns used:** State, City, Locality, Property_Type, BHK, Size_in_SqFt, Price_in_Lakhs, Year_Built, Furnished_Status, Floor_No, Total_Floors, Age_of_Property, Nearby_Schools, Nearby_Hospitals, Public_Transport_Accessibility, Parking_Space, Security, Amenities, Facing, Owner_Type, Availability_Status
- **Environment:** Google Colab, Python 3.13, scikit-learn 1.6.1, pandas 2.2.3, NumPy 2.1.3, XGBoost 3.4.1
- **Random seed:** 42 (used for all splits and models)

1.2 Data cleaning

Step	Action	Result
Duplicates	Checked all columns for duplicate rows	0 duplicates found
Identifier	Dropped the `ID` column	Not used as a feature
Missing values	Checked Price, Size and City for nulls	0 nulls found
Invalid values	Removed rows with Price or Size ≤ 0	0 rows removed
Outliers	Capped Price_in_Lakhs and Size_in_SqFt at the 1st and 99th percentile	Rows kept, extreme values limited
Type conversion	Converted numeric features with `to_numeric(errors="coerce")`	Safe handling of bad text values

After cleaning, the dataset has **250,000 rows and 22 columns**.

Capping was used instead of deleting rows so that no property records were lost. Price per sq ft was not capped separately. Section 2 (EDA) shows that about 7.9% of raw price-per-sq-ft values are high outliers, and they were kept in the data.

1.3 Feature engineering

- **Price_per_SqFt** = Price_in_Lakhs × 1,00,000 ÷ Size_in_SqFt. The Price_per_SqFt column in the original file is on a different scale, so it was recalculated.
- **Amenities_Count** = number of items in the comma-separated Amenities list.
- **Age_of_Property** was taken from the dataset (it matches Current Year − Year_Built).

- **Categorical encoding:** One-hot encoding for State, City, Property_Type, Furnished_Status, Public_Transport_Accessibility, Facing, Owner_Type, Availability_Status, Security and Parking_Space. Parking_Space and Security contain Yes/No values, so they are treated as categories and not as numbers.
- **Scaling:** StandardScaler is applied to numeric features for Logistic Regression and Linear Regression only. Tree-based models do not need scaling.

The brief also suggested a School Density Score. It was not created, because the dataset has only a count of nearby schools, so a density measure could not be calculated without area data.

1.4 Target variables

Classification target: Good_Investment

A property is labelled **1 (Good Investment)** if its price per sq ft is below the median price per sq ft of its own city. Otherwise it is labelled **0**.

- This is a domain rule chosen from the options in the brief (price vs median, price per sq ft vs median, multi-factor score).
- Class balance: 125,013 properties labelled 0 and 124,987 labelled 1 (about 50% each).

Regression target: Future_Price_5yr_Lakhs

Future price is calculated with the compound growth formula:

$$\text{Future_Price} = \text{Price_in_Lakhs} \times (1 + r)^5$$

The growth rate `r` depends on the city:

City	Annual growth rate (assumed)
Bangalore	10%
Mumbai, Hyderabad	9%
Pune, Ahmedabad	8%
Chennai, Delhi	7%
Kolkata	6%
All other cities	8% (base rate)

Important: The dataset does not contain real future prices, and the city growth rates are assumptions made for this project. The regression target is therefore a rule-based proxy, not a measured market outcome. This is discussed in Section 4 (Limitations).

1.5 Feature sets

- **Classifier features:** all numeric features and categorical features, **excluding** Price_in_Lakhs and Price_per_SqFt. These two columns were excluded because the label is calculated from price per sq ft. Including them would let the model copy the label rule instead of learning from other features.

- **Regressor features:** all numeric features (including Price_in_Lakhs) and categorical features. The regressor needs the current price, because the target is calculated from it.

1.6 Train/test split and preprocessing

- **Split:** 80% training (200,000 rows) and 20% test (50,000 rows), using `train_test_split` with stratification on Good_Investment.
- **Imputation:** Numeric missing values are filled with the median. Categorical missing values are filled with the most frequent value. This is done inside the pipeline, so the same steps run on new data.
- **Pipeline:** All preprocessing and the model are combined into one scikit-learn Pipeline, which avoids data leakage from test data into training.

1.7 Models

Classification (Good Investment)

Model	Settings
Logistic Regression	max_iter = 2000, scaled features
Random Forest	100 trees, max depth 15, min samples per leaf 5
XGBoost	400 trees, max depth 6, learning rate 0.05, subsample 0.8, column sample 0.8

Random Forest was trained on a random sample of 60,000 training rows to keep training time reasonable. The other models used all training rows.

Regression (5-year future price)

The model learns the **growth ratio** (future price ÷ current price), and the prediction is multiplied by the current price. This approach was chosen because tree-based models cannot predict values outside the range seen in training. Predicting a ratio keeps the output sensible for any price.

Model	Settings
Fixed-rate baseline	Every property grows at 8% per year (no learning)
Linear Regression	Scaled features
Random Forest	100 trees, max depth 15, min samples per leaf 5
XGBoost	500 trees, max depth 6, learning rate 0.05, subsample 0.8, column sample 0.8

1.8 Model selection and evaluation

- **Classification:** The best model is chosen by the **F1 score** on the test set. Accuracy, F1, ROC-AUC and the confusion matrix are reported.

- **Regression:** The best model is chosen by **RMSE** on the test set. RMSE, MAE and R^2 are reported. The fixed-rate baseline is included for comparison.

Note on model selection: The best model was selected using test-set results, because the project does not have a separate validation set. The reported test scores therefore slightly overstate how well the chosen model will generalise. This is listed in Section 4.

1.9 Experiment tracking

- All training runs were logged to **MLflow** under the experiment name `real_estate_advisor`.
- Logged items: model name, metrics (accuracy, F1, ROC-AUC, RMSE, MAE, R^2) and the fitted pipeline.
- The best classifier (Random Forest) and the best regressor (Random Forest) were registered in the **MLflow Model Registry** as `real_estate_classifier` (version 1) and `real_estate_regressor` (version 1).
- The MLflow log files are stored in `mlflow_logs/`, and can be viewed with: `mlflow ui --backend-store-uri sqlite:///mlflow_logs/mlflow.db`
- Earlier runs from the experiment history (for example, runs from before the regression ratio target and the Random Forest limits were introduced) are kept in the logs, but they are not used for the final results.

1.10 Output artifacts

File	Purpose
`models/classifier.joblib`	Saved Good Investment classifier (Random Forest pipeline)
`models/regressor.joblib`	Saved 5-year price regressor (Random Forest pipeline)
`reports/metrics.json`	All metrics, confusion matrix and top features
`data/app_sample.csv`	30,000-row sample used by the Streamlit app
`cleaned_housing_data.csv`	Full cleaned dataset (250,000 rows, 25 columns)
`mlflow_logs/`	MLflow experiment logs and registry

2. Findings (Exploratory Data Analysis)

This section answers the 20 EDA questions from the project brief. Each answer is based on the notebook outputs and charts, using the cleaned dataset (250,000 rows) unless stated otherwise.

Question numbers follow the project brief. Where the notebook used a different number, the brief numbering is used here.

2.1 Price and size analysis

Q1. What is the distribution of property prices? Prices range from about ₹15 lakh to ₹495 lakh, with a mean of ₹254.6 lakh and a median of ₹253.9 lakh. The middle 50% of properties lie between ₹132.6 lakh and ₹376.9 lakh. The histogram is almost flat across this range, meaning every price band has roughly the same number of properties. The taller bars at both ends come from capping extreme values at the 1st and 99th percentile during cleaning (see Section 1.2).

Q2. What is the distribution of property sizes? Sizes range from 545 to 4,956 sq ft, with a median of 2,747 sq ft. The middle 50% lies between 1,623 and 3,874 sq ft. As with price, the histogram is nearly flat, and the taller bars at the ends are the result of capping.

Q3. How does price per sq ft vary by property type? Apartments, Independent Houses and Villas have very similar price-per-sq-ft distributions. Their boxplots overlap almost completely, and the medians sit at about the same level (around ₹9,000 to ₹10,000 per sq ft). Property type does not appear to change the price per sq ft in this dataset.

Q4. Is there a relationship between property size and price? No clear relationship is visible. The scatter plot of size against price (sampled 5,000 points) forms a uniform cloud with no upward or downward trend. The correlation between size and price is close to zero. This is unusual for real estate data, where bigger homes normally cost more, and is discussed in Section 4.

Q5. Are there outliers in price per sq ft or property size?

Variable	Outliers (IQR method)	Share
Price per sq ft (raw data)	19,723	7.89%
Size (sq ft)	0	0%

All price-per-sq-ft outliers are on the high side, with values up to about ₹1,00,000 per sq ft. Size has no outliers. Outliers in price per sq ft were kept in the cleaned data, because they are a real part of the dataset and the Good Investment label is calculated from the median, which is not affected by them.

2.2 Location-based analysis

Q6. What is the average price per sq ft by state? Average price per sq ft is between about ₹12,900 and ₹13,200 in every state. Karnataka has the highest average and Rajasthan the lowest. The gap between the highest and lowest state is only about 2%, so state does not strongly change the price per sq ft in this data.

Q7. What is the average property price by city? Average price is also very similar across cities:

Group	Cities	Average price (lakh)
Top 15 (range)	Bangalore (highest), Surat, Kochi, Gaya, Mangalore and others	about 255 to 258
Bottom 5	Mumbai (252.1), Bilaspur (251.9), New Delhi (251.6), Trivandrum (251.3), Cuttack (250.8)	about 251 to 252

The total spread between the highest and lowest city is less than ₹7 lakh, which is less than 3% of the average price. Mumbai and New Delhi, which are among the most expensive cities in real life, appear at the bottom of this list. This is a clear sign that the dataset's city values do not follow real-market prices (see Section 4).

Q8. How is BHK distributed across cities? Each BHK type (1 to 5) makes up about 19% to 21% of properties in each of the top 10 cities. Overall there are about 50,000 properties per BHK type. No city has a dominant BHK type, which again differs from real markets, where cities usually have more 2 and 3 BHK homes than 5 BHK homes.

Q9. What is the median age of properties by locality?

- The dataset has 500 localities.
- Overall median property age is 18 years.
- Locality medians range from 16 years (Locality_443) to 21 years (Locality_72).
- The standard deviation of locality medians is only 0.78 years.

Locality has very little effect on property age. The youngest and oldest localities differ by only a few years.

Q10. What are the price trends in the top 5 most expensive localities? The top 5 localities by average price per sq ft are Locality_207, Locality_416, Locality_246, Locality_359 and Locality_387. Their median price per sq ft lies between about ₹9,800 and ₹10,400, and their median price between ₹257 and ₹267 lakh. The boxplots overlap almost completely.

The average-based ranking is pulled up by a small number of very high-priced properties (outliers up to ₹80,000 to ₹90,000 per sq ft). The medians are much closer together than the averages.

Limitation: The brief asks for price trends over time, but the dataset has no sale date. It only has `Year_Built`, which is when the property was constructed, not when it was sold. A time trend cannot be shown, so this question is answered with a comparison of price distributions instead.

2.3 Feature relationships and correlation

Q11. How are numeric features correlated with each other? The correlation heatmap shows the following:

- **Age_of_Property and Year_Built** are almost perfectly negatively correlated (-1). This is expected, because age is calculated from the year built.
- **Price_in_Lakhs and Future_Price_5yr_Lakhs** are almost perfectly correlated (close to 1). This is expected, because the future price is calculated from the current price.
- **Price_per_SqFt and Price_in_Lakhs** have a moderate positive correlation, because both use the same price value.
- **Size_in_SqFt and Price_per_SqFt** have a negative correlation. Larger properties have a lower price per sq ft, which matches the Good Investment result below.
- **Good_Investment and Size_in_SqFt** have a positive correlation. Larger properties are more likely to be labelled Good Investment.
- Most other features (BHK, Floor, Nearby_Schools, Nearby_Hospitals, Amenities_Count) have correlations close to zero with each other and with price.

Q12. How do nearby schools relate to price per sq ft? The median price per sq ft for properties with 1 to 10 nearby schools lies between about ₹9,140 and ₹9,350. The correlation between the number of nearby schools and price per sq ft is -0.0001, which means there is no measurable relationship.

Q13. How do nearby hospitals relate to price per sq ft? The median price per sq ft for properties with 1 to 10 nearby hospitals lies between about ₹9,130 and ₹9,350. The correlation is also -0.0001. Like schools, nearby hospitals do not affect price per sq ft in this data.

Q14. How does price vary by furnished status? Furnished, Semi-furnished and Unfurnished properties have very similar price distributions. Their boxplots overlap closely, and medians are around ₹250 lakh for all three. Furnishing status does not appear to change the price.

Q15. How does price per sq ft vary by facing direction?

Facing	Median price per sq ft (₹)
West	9,308
East	9,233
South	9,224
North	9,219

The gap between the highest and lowest direction is less than ₹100 (about 1%). Facing direction has no meaningful effect on price per sq ft.

2.4 Investment, amenities and ownership analysis

Q16. How many properties belong to each owner type? The three owner types have almost equal counts, at about 83,000 properties each (roughly one-third of the dataset).

Note on the brief: The brief lists owner types as Individual, Builder and Agent. The dataset actually uses **Owner, Builder and Broker**. The answer above uses the dataset values.

Q17. How many properties are available under each availability status?

- Ready_to_Move: about 125,000 properties
- Under_Construction: about 125,000 properties

The two statuses are split almost exactly in half.

Note on the brief: The brief lists Available, Under Construction and Sold. The dataset has only **Ready_to_Move** and **Under_Construction**. No property is marked as Sold or Available.

Q18. Does parking space affect property price?

Parking	Count	Median price (lakh)	Median price per sq ft (₹)
No	125,456	253.0	9,246
Yes	124,544	254.0	9,245

The difference is about ₹1 lakh in median price, and almost nothing in price per sq ft. Parking does not affect price in this dataset.

Note on the brief: The brief describes Parking_Space as a number of parking spots. In the dataset it only contains Yes or No, so the effect of the number of spots cannot be tested.

Q19. How do amenities affect price per sq ft? Median price per sq ft for properties with 1 to 5 amenities lies between about ₹9,206 and ₹9,294. The correlation between amenities count and price per sq ft is 0.002, which is effectively zero. Amenities do not change price per sq ft in this data.

Q20. How does public transport accessibility relate to price per sq ft and investment potential?

Transport access	Median price per sq ft (₹)	Good Investment rate	Properties
Low	9,265	49.9%	83,287
Medium	9,269	49.8%	83,008
High	9,205	50.3%	83,705

Transport access has almost no effect on price per sq ft or on the share of Good Investment properties. The Good Investment rate is about 50% in every group, which is expected, because the label was created from a median split (see Section 1.4).

2.5 Summary of key findings

1. **Price and size are both almost uniformly distributed.** The dataset does not show the usual real-estate pattern where larger homes cost more.

2. **Location (state, city, locality) has very little effect** on price per sq ft or average price.
3. **Most property features have no measurable effect** on price per sq ft. These include schools, hospitals, facing direction, furnishing, parking, amenities and transport access.
4. **Size is the only feature with a clear relationship** to the Good Investment label, because larger properties have a lower price per sq ft. The ablation test (Section 3.1) confirms this: without size, the classifier performs at chance level.
5. **Locality ages are very similar** (median 16 to 21 years).

2.6 Data quality notes

These points should be stated in the report, because they affect how the findings should be read:

- The distributions of price, size, BHK, owner type and availability are almost perfectly flat. Real market data is usually skewed, so the dataset may be simulated or heavily processed.
- Several categories in the brief do not match the dataset (owner types, availability statuses, parking format).
- Crime rate and infrastructure score, which the brief asks to be analysed, are not in the dataset. Proxy features (Security, Public Transport, Nearby Schools, Nearby Hospitals) were used instead.
- Nearby_Schools and Nearby_Hospitals contain only the whole numbers 1 to 10, and Amenities_Count only 1 to 5.

3. Model Performance

This section reports how the classification and regression models performed on the held-out test set (50,000 properties, 20% of the data). All numbers come from `reports/metrics.json` and the notebook outputs. The models were trained on the remaining 200,000 properties.

3.1 Classification: Good Investment

Comparison of models

Model	Accuracy	F1 score	ROC-AUC
Logistic Regression	0.7081	0.7071	0.7794
Random Forest (selected)	**0.7083**	**0.7093**	0.7729
XGBoost	0.7075	0.7086	0.7774

Selection rule: the model with the highest F1 score was selected. Random Forest was selected with an F1 score of 0.7093.

All three models perform almost identically (accuracy between 70.75% and 70.83%, F1 between 0.707 and 0.709). The differences are too small to show that one model is truly better than the others. Logistic Regression has the highest ROC-AUC (0.7794), which means it ranks properties slightly better, but it was not selected because the selection rule was based on F1.

Classification report (Random Forest, test set)

Class	Precision	Recall	F1 score	Support
Not good (0)	0.71	0.70	0.71	25,003
Good (1)	0.71	0.71	0.71	24,997
Overall accuracy			**0.71**	50,000

Precision and recall are balanced between the two classes, so the model does not favour one label over the other.

Confusion matrix (Random Forest, test set)

	Predicted: Not good	Predicted: Good
Actual: Not good	17,625 (correct)	7,378 (wrong)
Actual: Good	7,207 (wrong)	17,790 (correct)

- Total correct predictions: 35,415 out of 50,000 (70.8%).
- Total wrong predictions: 14,585 (29.2%).
- Errors are split evenly between the two directions, so the model does not show a bias towards one label.

Interpretation

- A random guess on this balanced data would be right about 50% of the time. The model is right about 71% of the time, which is a real improvement, but not a strong one.
- The ceiling is limited by the labelling rule. The label depends on each property's price per sq ft relative to its city median. The model cannot see price columns, so it has to estimate this from other features, and it does so with limited accuracy.
- Accuracy around 70% is a reasonable result for this type of problem. It should not be presented as a reliable investment advisor. The model is best described as a tool that highlights which properties are likely to be cheap for their city, based on size, location and other features.

Feature importance (Random Forest classifier)

Rank	Feature	Importance
1	Size_in_SqFt	0.677
2	Floor_No	0.027
3	Total_Floors	0.027
4	Year_Built	0.023
5	Age_of_Property	0.023
6	Nearby_Hospitals	0.019
7	Nearby_Schools	0.018
8	Amenities_Count	0.014
9	BHK	0.014
Rest	Parking, Security, Owner type and others	below 0.006 each

Size_in_SqFt accounts for about 68% of the total importance. Every other feature has a small share. This is the most important finding of the classification model. Because price per sq ft falls as size rises in this dataset, the model mostly predicts "Good Investment" for larger properties and "Not good" for smaller ones. Section 4 explains why this is a limitation and not a valid investment rule.

Ablation test: classifier without Size_in_SqFt

To check whether the classifier learns anything beyond size, all three classifiers were retrained with `Size_in_SqFt` removed. Everything else stayed the same.

Model	With size: accuracy / F1	Without size: accuracy / F1
Logistic Regression	0.7081 / 0.7071	0.4956 / 0.4915
Random Forest	0.7083 / 0.7093	0.4972 / 0.5221
XGBoost	0.7075 / 0.7086	0.4972 / 0.4986

Result: Without size, accuracy drops to about 50%, which is the same as random guessing on this balanced dataset. The F1 scores of 0.49 to 0.52 also show no real predictive power. This means that **almost all of the classifier's accuracy comes from Size_in_SqFt.** The other features (floor, age, year built, nearby schools and hospitals, amenities, BHK, parking, security, owner type) contribute nothing measurable to the Good Investment label.

Why this happens: In this dataset price is almost independent of size (Section 2, Q4). Price per sq ft is price divided by size, so for a roughly fixed price, a larger size gives a lower price per sq ft. The label is "price per sq ft below the city median", so size alone is enough to predict it. This is a property of the data and the label rule, not evidence that larger homes are better investments.

3.2 Regression: 5-year future price

Comparison of models

Model	RMSE (lakh)	MAE (lakh)	R^2
Fixed-rate baseline (8% per year for all)	9.55	2.59	0.9979
Linear Regression	0.0002	0.0001	1.0000
Random Forest (selected)	**0.0000**	**0.0000**	**1.0000**
XGBoost	0.0078	0.0024	1.0000

Selection rule: the model with the lowest RMSE was selected. Random Forest had the lowest RMSE.

How to read these results

The R^2 values close to 1.0 and the very low errors **do not show that the models predict the real market well.** The dataset has no actual future prices. The target was calculated with a fixed formula:

Future price = Current price × (1 + city growth rate)^5

The city growth rates are assumptions made for this project (see Section 1.4). The models learned the growth rate for each city from the data and then reproduced the formula almost exactly. This is why the errors are so small.

The more useful comparison is between the **fixed-rate baseline** and the **ML models**:

- The fixed 8% rule, which ignores city, has an RMSE of 9.55 lakh.
- The ML models, which learned the city-specific rates, have an RMSE close to zero.

This result shows that the models can recover a pattern that is built into the data. It does not show that they can forecast real house prices.

Feature importance (Random Forest regressor)

Rank	Feature	Importance
1	City: Kolkata	0.413
2	City: Bangalore	0.301
3	City: Hyderabad	0.111
4	City: Chennai	0.103
5	City: Mumbai	0.072

City features make up almost all of the importance. This is expected, because the growth rate is set per city. Price and size have almost no importance, because the model only needs to learn the growth ratio, not the price itself.

3.3 Model selection summary

Task	Selected model	Main metric	Score on test set
Good Investment (classification)	Random Forest	F1 score	0.7093
5-year price (regression)	Random Forest	RMSE	≈ 0 lakh (rule-based target)

Both selected models are Random Forests. They were trained with the same settings (100 trees, max depth 15, minimum 5 samples per leaf). The Random Forest classifier was trained on a sample of 60,000 rows, and the regressor on the same sample, to keep training time reasonable.

3.4 Overfitting check

Overfitting means the model performs much better on the training data than on new data. The notebook includes a check that compares training and test scores.

Measure	Training score	Test score	Gap
Classifier F1 (Random Forest)	0.7203	0.7093	0.0110 (about 1.1 points)
Regressor R^2 (Random Forest)	1.0000	1.0000	0.0000

A gap of 2 to 3 F1 points or less is generally considered acceptable. A larger gap would indicate overfitting.

Result:

- **Classifier:** The training F1 (0.7203) is about 1.1 points higher than the test F1 (0.7093). This is within the acceptable range, so the Random Forest shows only mild overfitting. The model is learning patterns that generalise to new data, although it does memorise the training set slightly more than it does the test set.

- **Regressor:** The R^2 is 1.0 on both training and test sets. This is expected, because the target is a fixed formula. It does not indicate overfitting, but it also does not show real forecasting skill (see Section 3.2).

The training score was measured on all 200,000 training rows, while the Random Forest was fitted on a 60,000-row sample. Because the evaluation rows include rows the model did not see during fitting, the gap reported here is a fair measure of generalisation.

3.5 Model artifacts

- The trained models are saved as `models/classifier.joblib` and `models/regressor.joblib`.
- Both models are registered in the MLflow Model Registry (see Section 1.9).
- The full set of metrics is in `reports/metrics.json`.
- The Streamlit app loads these saved models to make predictions for new properties.

4. Limitations

This section lists the main limitations of the data, the targets, the models and the app. They should be read together with the results in Sections 2 and 3. Several results that look strong (such as $R^2 = 1.0$) are only strong because of these limitations.

4.1 The dataset looks synthetic

Several patterns in the data do not match what is normally seen in real estate markets:

- **Flat distributions:** Price, size and BHK are almost evenly spread across their ranges. Real housing prices are usually skewed, with more properties in the lower and middle price bands.
- **No price-size relationship:** Larger properties do not cost more. The scatter plot of size against price shows no trend (Section 2, Q4).
- **Identical city prices:** The average price of the top 15 cities differs by less than ₹7 lakh. Mumbai and New Delhi, which are among the most expensive cities in India, appear at the bottom of the list (Section 2, Q7).
- **Even BHK split:** Each BHK type makes up about 20% of properties in every city. Real cities have many more 2 and 3 BHK homes than 5 BHK homes (Section 2, Q8).
- **Small discrete features:** Nearby_Schools only takes values 1 to 10, Nearby_Hospitals 1 to 10, and Amenities_Count 1 to 5.
- **Equal category counts:** Owner types, availability statuses and parking are almost exactly split in half or into thirds.

Impact: Findings from this dataset describe the dataset, not the real Indian housing market. Conclusions such as "parking does not affect price" or "schools do not affect price" should not be taken as real-world facts.

4.2 The Good Investment label is a rule, not an outcome

- The label is defined as "price per sq ft below the median of the city". It is not based on rental yield, resale value, or what buyers actually earned.
- Because the label is defined this way, about 50% of properties are "Good" by construction.
- The classifier is learning to reproduce this rule from other features. Its accuracy (about 71%) reflects how well those features predict the rule, not how well they predict investment success.

4.3 The 5-year future price is a formula, not a forecast

- The dataset does not contain real future prices. The 5-year price was calculated with a compound growth formula.
- The city growth rates (for example, Bangalore 10%, Kolkata 6%) are assumptions made for this project, not measured data.
- Real property prices depend on many factors that are not in the model, including interest rates, inflation, local supply and demand, government policy and economic cycles.

- The R^2 value of 1.0 reflects how well the models reproduce the formula. It does not show accuracy on real future prices.

Impact: The regression models should not be used to predict actual future prices. They are a demonstration of how the pipeline works, and the growth rates could be replaced with real historical data if it becomes available.

4.4 The classifier depends mainly on size

- Size_in_SqFt accounts for about 68% of the classifier's feature importance (Section 3.1), and the ablation test confirms that the classifier depends on it almost entirely.
- In this dataset, price per sq ft falls as size rises. The model therefore learned "larger properties are more likely to be Good Investments."
- This is a pattern in the data, not a sound investment principle. In real markets, larger homes often have a lower price per sq ft, but that does not make them better investments.

Impact: The app should not be used to recommend larger properties as better investments. This was tested (Section 3.1, ablation test). With Size_in_SqFt removed, accuracy drops to about 50%, which is random guessing. The classifier's performance is therefore almost entirely explained by size, and the other features add no measurable predictive value.

4.5 Missing features

The following features were mentioned in the project brief but are not available in the dataset:

Feature in brief	Status	How it was handled
Crime rate	Not in dataset	Not used. Security (Yes/No) used as a rough proxy
Infrastructure score	Not in dataset	Not used. Public transport access and nearby schools and hospitals used as proxies
School density score	Not in dataset	Not created (needs area data)
Sale date	Not in dataset	Time trends cannot be measured (Section 2, Q10)
Locality coordinates	Not in dataset	Location analysis uses locality names only
Real future prices	Not in dataset	Future price is calculated from a formula
Rental income	Not in dataset	Investment return is not measured

4.6 Category mismatches with the brief

- **Owner types:** The brief lists Individual, Builder and Agent. The dataset uses Owner, Builder and Broker.

- **Availability status:** The brief lists Available, Under Construction and Sold. The dataset only has Ready_to_Move and Under_Construction.
- **Parking:** The brief describes the number of parking spots. The dataset only has Yes or No.

These differences were handled by using the dataset values. The brief's questions about these features can only be answered for the categories that exist in the data.

4.7 Model selection and evaluation

- **No separate validation set:** The best model was selected using the test set. This means the reported test scores are slightly optimistic, because the choice of model was influenced by the same data used to report performance. A three-way split (training, validation, test) would avoid this.
- **Sampling for Random Forest:** Random Forest was trained on a random 60,000-row sample of the 200,000 training rows, to keep training time reasonable. This may slightly reduce its accuracy compared with training on all rows.
- **Small differences between classifiers:** The three classifiers differ by less than 0.5 percentage points in accuracy. The choice between them is not strongly supported by the results.
- **Fixed decision threshold:** The classifier uses a probability of 0.5 as the cutoff for "Good". This cutoff was not tuned, and a different cutoff might give different precision and recall.
- **Overfitting check:** The comparison of training and test scores is recorded in Section 3.4. If the gap is large, the model may be memorising the training data rather than learning general patterns.

4.8 Data cleaning effects

- **Capping outliers:** Price and size were capped at the 1st and 99th percentile. This removes the most extreme values and makes the distributions look more uniform than the raw data. The capping may hide real high-end and low-end markets.
- **Price per sq ft outliers kept:** About 7.9% of properties have unusually high price per sq ft values (Section 2, Q5). These were kept in the data, which means averages are pulled upward by these values. Medians are used where possible for this reason.
- **Growth rate defaults:** Only 8 cities have a specific growth rate. All other cities use the default rate of 8%, which means most properties in the dataset use the same rate.

4.9 Deployment and app limitations

- **Sample data in the app:** The Streamlit app uses a random 30,000-row sample of the data for the price comparison and the Explore tab. The models themselves were trained on all 250,000 rows, but the app's peer comparison is based on the sample only.
- **Library version pinning:** The app's requirements are pinned to the versions used in training (scikit-learn 1.6.1, XGBoost 3.4.1, pandas 2.2.3). If the environment changes, saved models may not load correctly.
- **No live data:** The app does not connect to any live property listing or market data source.

- **No user feedback:** The app does not record user inputs or check whether predictions turned out to be correct.

4.10 Scope of use

This project is an academic demonstration of a data science pipeline. It should not be used as:

- Financial advice, or a basis for buying or selling property
- A valuation tool for real transactions
- A replacement for professional valuation by a licensed property valuer
- A source of decisions about specific neighbourhoods, cities or groups of people

Any real-world use would require real transaction data, validation against actual outcomes, and review by domain experts.

5. Use Cases and Business Impact

This section describes how the project is intended to be used, which users it serves, and what business value it can and cannot provide. The use cases come from the project brief. For each one, this section states what the system does, how it helps, and what limits its usefulness.

5.1 Overview

The brief defines four business use cases:

1. Empower real estate investors with intelligent tools to assess long-term returns.
2. Support buyers in choosing high-return properties in developing areas.
3. Help real estate companies automate investment analysis for listings.
4. Improve customer trust in real estate platforms with data-backed predictions.

The Streamlit application provides three features that support these use cases:

App feature	What it does
Predict tab	Analyses one property. Returns whether it is a Good Investment (with a confidence score), the estimated price after 5 years, the expected gain, and a comparison with other properties in the same city and BHK category.
Explore tab	Filters the data by state, BHK and price. Shows average price per sq ft by city, the Good Investment rate by property type, and a heatmap of price per sq ft by city and BHK.
Model Performance tab	Shows the metrics, confusion matrix, feature importance and EDA charts for transparency.

5.2 Use case 1: Investors assessing long-term returns

Who: Individual investors and small investment groups who are deciding whether a property is worth buying to hold for several years.

How the system helps:

- The Predict tab gives a quick first screen. An investor can enter the details of a property and see whether it is cheap for its city, based on price per sq ft.
- The 5-year price estimate gives a simple view of possible value growth, using city-level growth rates.
- The peer comparison shows whether the property's price per sq ft is above or below the median for its city and BHK category.

Limits:

- The Good Investment label is based on price per sq ft relative to the city median. It does not measure rental income, resale history, or actual returns.
- The 5-year price uses assumed growth rates, not historical market data (see Section 4.3). It should not be used as a forecast.
- The classifier is correct about 71% of the time on test data. It should be treated as a screening aid, not a decision tool.

Appropriate use: The app can help an investor shortlist properties for further research. It should always be used with local market knowledge, legal checks and professional advice.

5.3 Use case 2: Buyers choosing high-return properties in developing areas

Who: Home buyers who want to find properties that may gain value, especially in areas that are still developing.

How the system helps:

- The Explore tab allows filtering by state, BHK and price, so a buyer can compare price levels across cities.
- The heatmap of price per sq ft by city and BHK helps identify cities where a given property size is relatively cheap.

Limits:

- The dataset has no information about whether an area is developing. There is no infrastructure score, no planned-development data, and no information about new transport links or business parks. The brief mentions these factors, but they are not in the data (see Section 4.5).
- The Explore tab does not filter by locality, so a buyer cannot compare neighbourhoods within a city.
- The dataset shows almost no difference in price per sq ft between cities or localities (Section 2, Q6 to Q10). The data cannot show which areas are cheaper or growing faster.

Appropriate use: The app can show whether a specific property is priced low for its city. It cannot identify developing areas. This use case needs additional data on infrastructure, planned projects and historical price growth before it can be supported.

5.4 Use case 3: Real estate companies automating investment analysis for listings

Who: Property portals, brokerages and developers who publish many listings and want to tag them with an investment indicator.

How the system helps:

- The trained classifier and regressor are saved as files (`models/classifier.joblib` and `models/regressor.joblib`) and can be loaded in any Python environment.

- The same prediction function used in the app (`predict_property`) can be run in a loop over a list of listings to produce a Good Investment flag and a 5-year price estimate for each one.
- Batch processing takes only a few seconds per thousand listings on a standard machine, based on the prediction speed of the Random Forest models.

Limits:

- Listings need the same input fields as the training data, including Price, Size, City, BHK, Year_Built and Amenities. Missing fields are filled with median or most-frequent values, which may reduce accuracy.
- Category values in the listings must match the dataset categories (for example, Owner_Type values Owner, Builder and Broker). Other values are treated as unknown.
- The model reflects the patterns in this dataset. If a company's listings come from a different market, the predictions may not be reliable without retraining on that company's data.

Appropriate use: A company could use the model to flag listings for internal review, for example to highlight properties priced low relative to their city. It should not use the flag as the only basis for pricing or for advice to customers.

5.5 Use case 4: Improving customer trust in real estate platforms

Who: Online real estate platforms and their customers. Buyers and renters look at listings on these platforms and want to know whether the information shown is reliable. The platform's product and trust teams want to add data-backed insights without misleading users.

How the system helps:

- **Transparent indicators:** The Predict tab shows the price per sq ft against the city and BHK median, so a user can see the reasoning behind a price signal, not just a verdict.
- **Visible confidence:** The Good Investment result shows a probability (for example, 34% or 58%), not just "Yes" or "No". Users can see how sure the model is.
- **Explained drivers:** The Model Performance tab shows which features the model relies on. This makes clear that size is the main driver, which helps users judge the result.
- **Published accuracy:** The confusion matrix, F1 score and ablation results are available in the app and documentation. A platform can show these numbers, which builds trust through openness.
- **Disclaimer:** The app states that it is for learning and screening, not financial advice.

Limits:

- Trust built on data only lasts if the data is accurate. The classifier is correct about 71% of the time, and the 5-year price uses assumed growth rates (see Section 4).
- The most important finding is that the Good Investment result depends almost entirely on size (Section 3.1). If a platform presented this result as a general quality score for properties, it would mislead users and damage trust once they noticed.

- The R^2 of 1.0 on the regression is not a sign of accuracy. Showing it as proof of accurate forecasting would be dishonest, so the platform must explain it correctly.

Appropriate use: Use the model as one input among many. Always show the confidence score, the main drivers and the limitations next to any prediction. Do not show a single "Good Investment" badge without context. Platforms should run a review with real sales data before using the predictions publicly.

Trust risk: Overclaiming is the main risk. If the platform presents the 5-year price as a forecast, or the Good Investment label as a measure of quality, users may rely on it for large financial decisions. Clear labelling of assumptions and limits is necessary for trust to last.

5.6 Business impact assessment

The project was built as an academic demonstration. The following table summarises the potential impact, and what would be needed to measure it properly.

Area	Potential impact	Evidence available now	What is needed to measure it
Investor screening	Faster first-pass review of many properties	Predict tab works, classifier ~71% accurate on test data	User testing with real investors, comparison with expert screening
Buyer decisions	Quick check of whether a price is high or low for a city and BHK	Peer comparison in Predict tab, Explore heatmap	Real market price data, and evidence that buyers use the tool
Company automation	Automatic tagging of listings	Batch-ready model files, prediction function	Trial on real listings, measurement of time saved and error rate
Return assessment	Estimate of long-term value	5-year estimate using assumed growth rates	Real historical price data for each city and locality
Customer trust	Data-backed, transparent predictions on listing platforms	Confidence score, feature importance, published metrics and disclaimer in the app	User studies on whether users understand the limits, and an audit against real sales outcomes

Summary of business value: In its current form, the project shows that a simple, explainable model can be built and deployed. It can support early screening of properties. Its main business value would come from using real market data, which is the most important step before any commercial use.

5.7 Risks and responsible use

Risk	Description	Mitigation
Over-reliance on the label	Users may treat "Good Investment" as financial advice	Clear disclaimer in the app and documentation (see Section 4.10)
Bias toward larger properties	The classifier depends on size (Section 4.4)	Document the dependence, and test the model without size before any real use
False growth expectations	The 5-year estimate may be read as a real forecast	Label the output as based on assumed growth rates
Errors on real data	Model accuracy on real listings is unknown	Validate on real listings with known outcomes before use
Data privacy	Real listings may contain personal information	Use only property features, not seller or buyer details
Loss of trust	Users discover the model's limits after relying on it	Show confidence, drivers and limitations next to every prediction

5.8 Recommended next steps

If this project were developed further for real use, the following steps would be needed, in order of priority:

1. **Get real market data:** historical sale prices by city and locality, with sale dates.
2. **Add the missing features:** infrastructure, planned development, crime data and transport links, as described in the brief.
3. **Replace the rule-based labels:** define Good Investment using actual returns (for example, price growth over several years after accounting for costs).
4. **Redefine the label:** the size ablation test (Section 3.1) showed the current label is effectively a size rule. A useful label needs real return data, or a price measure that is not tied to size.
5. **Validate with a three-way split:** separate training, validation and test sets, to avoid test-set selection bias.
6. **Get expert review:** have a property analyst or economist review the features, labels and assumptions.

These steps would turn the academic prototype into a tool that could support real investment decisions.